

Next-Generation Cloud Incident Responses for Military Networks

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Abstract—This paper explores an automated incident response approach for military networks hosted on cloud infrastructure, utilizing Infrastructure-as-Code (IaC) principles to enable rapid and scalable recovery mechanisms. The methodology includes isolating compromised components, preserving evidence for forensic analysis, and rebuilding affected resources to ensure the continuity of critical military services. The proposed approach provides a resilient, automated, and scalable solution to maintain critical service availability during cyber incidents. The experimental results are analyzed to validate the effectiveness of the methodology in real-world military network scenarios. The paper details the challenges faced while executing incident response in the cloud. It also outlines the possibility for future improvements by integrating autonomous cyber defense agents, automating incident detection and response triggers, implementing decoys, and transforming the environment into a continuous training ground for the red team. This paper was originally presented at the NATO Science and Technology Organization Symposium (ICMCIS) organized by the Information Systems Technology (IST) Scientific and Technical Committee, IST-209-RSY – the ICMCIS, held in Oeiras, Portugal, 13-14 May 2025.

Index Terms—Cloud Incident Response, Military Networks, Tactical Edge

I. INTRODUCTION

NATO and coalition members increasingly adopt cloud systems to enhance scalability, operational flexibility, and collaboration across geographically dispersed environments. At the same time, due to geopolitical dynamics, military networks are exposed to an increasing number of threats. These conditions raise the critical need for innovative and robust incident response mechanisms tailored to the unique operational and security requirements of NATO and allied military networks.

The cloud incident responses differ from the incident responses executed in on-premise infrastructures. Cloud environments enhance incident responses by providing the ability to scale response resources, easy automation, and built-in advanced security tools. Infrastructure-as-Code (IaC) principles implemented in the cloud services open the possibility to rapidly increase the reaction time, flexibility, and automation of each response. This paper focuses on incident response in military-owned cloud infrastructure. In contrast to the shared responsibility model used in public cloud computing, in a military cloud infrastructure,

all responsibilities related to incident responses must be accounted for by the military personnel managing the infrastructure and cannot be shared with the Cloud Service Provider (CSP). While the CSP provides essential security services and tools, military cloud incident response must remain under the direct control of military defense operators to detect, respond to, and mitigate threats within their cloud environment. While CSP offers foundational security measures, they are not directly involved in mission-specific operations. CSP are not involved in the situational awareness of the military operation and cannot take part of the decision making process in such contexts.

In our work, we assume that before the compromised events triggered the responses, CSP and the customer had already taken all required actions to mitigate the pre-incident risk. These actions include ensuring proper logging, monitoring, and boundary protection.

In our research scenario, despite all the actions taken to prevent a cyber security incident, there is evidence of a breach, and the containment phase of the incident response plan is activated. Previous research on this topic highlights the evolving area of cloud incident responses [1], [2], where traditional methods are being adapted and supplemented with cloud-native approaches to address specific challenges [3]. Incident response models such as National Institute of Standards and Technology (NIST) Special Publication (SP) 800-61 [4] and organizations such as Cloud Security Alliance (CSA) [5] are developing comprehensive frameworks to unify cloud incident response practices. These frameworks aim to provide guidance tailored to cloud users and service providers, addressing both technical and operational aspects of incident response. Our research fits into and combines the third phase of the incident response model of NIST SP 800-61 (Revision 2), 'Containment, Eradication and Recovery' and the elements 'Preparation' and 'Automation and orchestration' of the CSA approach [6] to provide a specialized framework for managing incidents in cloud environments.

The structure of the rest of the paper is as follows. Section II details the tools used to build and test the infrastructure and the definitions established for various response types. Section III outlines the design of responses and how they relate to adversarial tactics and techniques. Section IV describes the challenges encountered during the

research, mainly those unique to cloud environments, and explores the distinctions in risk management and decision-making processes between military and industry-owned contexts. It also provides an overview of potential paths for future research that could expand on the findings of this study. The last part of the document, Section V presents the results and findings of our investigation.

II. BACKGROUND AND DEFINITIONS

In our research, we have used a virtual laboratory provided by the NATO Allied Command Transformation (ACT) Science and Technology Experimentation Environment (SnTEE), which supports the transfer of scientific and technical activities carried out by the NATO Science and Technology Organization (STO) research task groups to NATO operations [7]. To build, change and version the infrastructure on Amazon Web Services (AWS), we used Terraform scripts executed through the Continuous Integration & Continuous Delivery (CI/CD) pipeline provided by the virtual laboratory. Incident responses are assembled from scripts written in Terraform, Python, and AWS Command Line Interface (CLI) combined by Ansible playbooks.

A. Network Topology

The chosen infrastructure was inspired by the network topology used in Cyber Operations Research Gym (CyBORG) Cyber Autonomy Gym for Experimentation (CAGE) challenge 4 [8], adapted to a cloud-based infrastructure. It consists of three Availability Zones (AZs) within a Virtual Private Cloud (VPC). In two of the AZs, the subnets host only mission operational Elastic Compute Cloud (EC2) instances. The third AZ hosts an unclassified but mission-critical military application. From a networking perspective, the environment includes route tables in each subnet, an Internet gateway, and a firewall on the edge of the VPC border. In addition, the private subnet hosting the military application contains a database instance, labeled as Amazon RDS in Fig. 1, which serves as the application's repository.

The mission-critical unclassified application implemented in the infrastructure is a real operational tool called Cyber Common Operational Picture (CyCOP). It is a Cyber Hybrid Situational Awareness (CyHSA) tool developed by the Technical University of Valencia in cooperation with the Spanish Joint Cybercommand and Spanish Ministry of Defense [9]. The application is hosted on three EC2s within the public subnet of the AZ. The availability of this application during the execution of the incident response is measured as one of the Key Performance Indicatorss (KPIs) for the effectiveness of the response. CyCOP core function is to integrate data from traditional cyber defense systems, such as Security Information and Event Management (SIEM), sensors, and Intrusion Detection System (IDS), with georeferenced physical information and provide a unified decision-making environment. The advanced visualization capabilities of CyCOP allow dynamic transitions between strategic, operational, and tactical perspectives and support

decision-making across different domains. By incorporating information from land, sea, air, space, and cyberspace, CyCOP offers a comprehensive view of hybrid operational environments.

B. Hierarchy of Responses

To provide integration with the multi-level concept of autonomous cyber agents [10]–[12], proposed by the North Atlantic Treaty Organization (NATO) Information Systems Technology (IST)-162 and IST-196 research groups, we decided to separate incident responses into four vertical levels: Instance response, Subnet response, Software-Defined Networking (SDN) response, and AZ response. Each incident response performs tasks to isolate, contain, and replace compromised instances within its operational level. The main objectives are to provide incident containment and recovery while ensuring the availability of mission-critical operations. Each higher-level response envelopes the activities provided by the lower-level responses, as shown in Fig. 2 and Fig. 3. The Terraform scripts used for responses are collected in a Git repository. Each relies on images or baselines that the infrastructure administrators can modify. Thus, engineers can change the EC2 baselines to avoid reintroducing the incident into the infrastructure after executing the response.

1) *Instance-level response*: This is the lowest level of response in which only a single EC2 instance is affected. The response is triggered once a detection mechanism categorizes the instance as compromised. It aims to prevent a security incident from spreading and eliminate a potentially established adversary foothold on the instance while immediately re-creating it and preserving evidence for future forensic analysis. The first action of the response is to take a snapshot of the compromised instance. This snapshot will provide possibilities for forensic investigations of the tools and methods used by adversaries. The second action will destroy and remove the compromised object from the infrastructure. The Instance-level response will re-create the object using clean configuration baselines and verified images, ensuring its fully restored network accessibility and functionality.

2) *Subnet-level response*: At this level, the response will take action on all instances within a specific subnet. This response should be triggered in case a specific number of instances in a subnet are categorized as compromised, or a threshold for executing the lower-level response has been met. The actions are similar to the lower-level response but on a larger scale. Snapshots will be taken of all instances for further forensic analysis. All instances will be destroyed and re-created to provide infrastructure resilience and availability. The IaC principles on this level offer a rare opportunity for post-incident analysis, as using the snapshot analyst could potentially re-create the affected network segment in a sandbox environment and conduct a detailed analysis of the adversary's tools and techniques.

3) *AZ-level response*: This level of response is the most global introduced in the research. It affects all instances in all

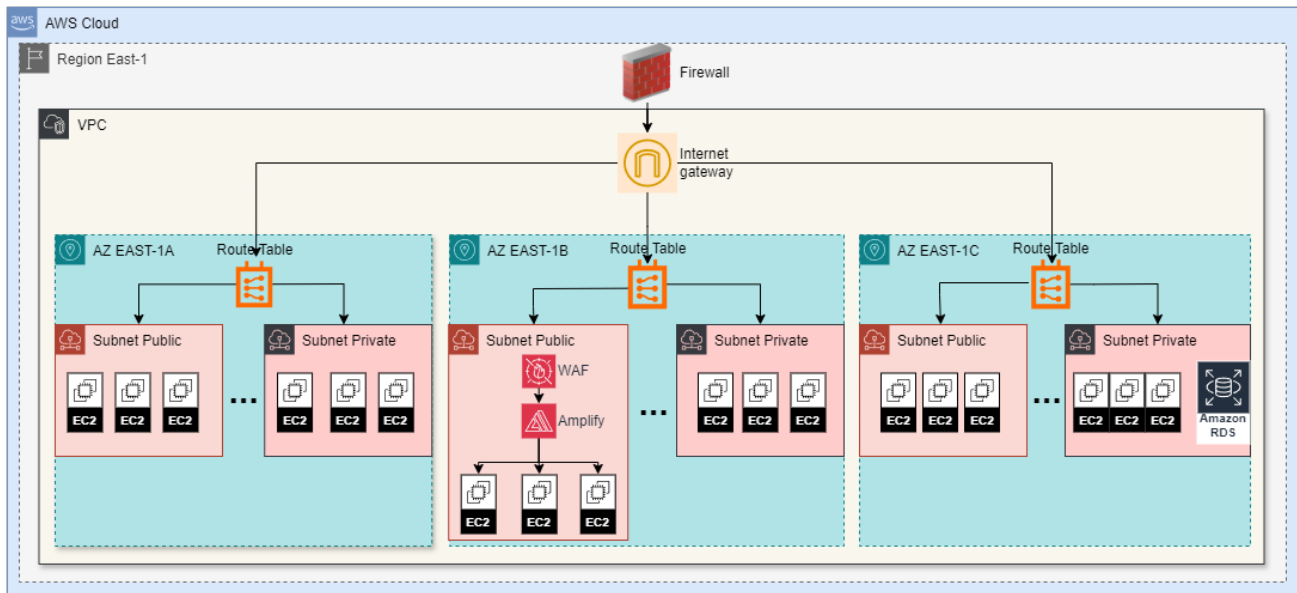


Fig. 1: Network topology describing the cloud experimental environment built on Amazon Web Services

subnets defined within an AZ. It follows the workflow of the lower-level responses by taking a snapshot and destroying and re-creating the instances. Recreating actions at this level will rebuild the affected infrastructure in a different AZ and ensure the usage of a different geographically distributed data center to provide additional protection against distributed attacks.

4) *SDN-level response*: The SDN-level response is the most sophisticated response that we have introduced in our environment. It aims to safeguard the infrastructure hosting military-critical applications in the cloud environment when there are indications that some or all components of that infrastructure may be compromised. In that case, the response, which includes preserving the evidence by creating a snapshot and destroying the entire infrastructure, should also ensure that the mission-critical infrastructure is broadly accessible by the other endpoints in the network. Once the infrastructure platform is restored to a new location, all internal network routing is reconfigured to restore the accessibility of the mission-critical application from all other parts of the cloud environment. This action should minimize service downtime while providing the same quality of protection as other responses.

III. RESPONSE DESIGN

The design of the responses is shaped according to the characteristics we identified as integral parts of cloud infrastructures for military use. These characteristics include how incident response decisions are made, the efficiency of responses, and how to measure their performance to improve them. To map the responses to real attack vectors, we create a mapping between them and a matrix of real-world tactics and techniques used by cyber adversaries. We investigated the possibility of implementing incident response, as described in this paper, in different topologies as described in the following subsections.

A. Decision making

The level of response executed at any given time is determined by the specific number of Indicators of Compromise (IOC) received by the triggering mechanism and the roles of the affected EC2s. The use of scripted responses allows us to consider different approaches, such as relying on a manual trigger performed by human operators in a Security Operations Center (SOC); using a centralized orchestrating tool that relies on predefined workflows; or integrating autonomous cyber-defense agents that can trigger responses when detecting abnormal behavior. In the following examples, we discuss the advantages and disadvantages of these options and suggest a universal decision-making framework.

1) *SOC with human operators*: In a SOC with human operators in place, decision making should combine automated alerting systems with human analysis. The alerts generated by monitoring tools, e.g., SIEM, IDS, Network Traffic Analysis (NTA), and log analyzer tools, are initially assessed by the SOC analysts. The analysts evaluate the incident and propose an appropriate response. This approach benefits from the possibility of human operators adding context and situational awareness when analyzing an incident, such as intelligence reports, mission-critical objectives, or recent events, to choose and guide the correct response. This ensures that actions are aligned with operational goals. The drawback of this approach is the slow response speed due to the required manual analysis. It requires skilled personnel and very well-implemented monitoring. In addition, it allows for human errors and delayed decisions.

2) *Using an orchestration tool*: This approach involves using orchestration tools to automate decision-making processes. These tools combine inputs from different monitoring systems and rely on predefined workflows to execute responses based on specific triggers. This approach

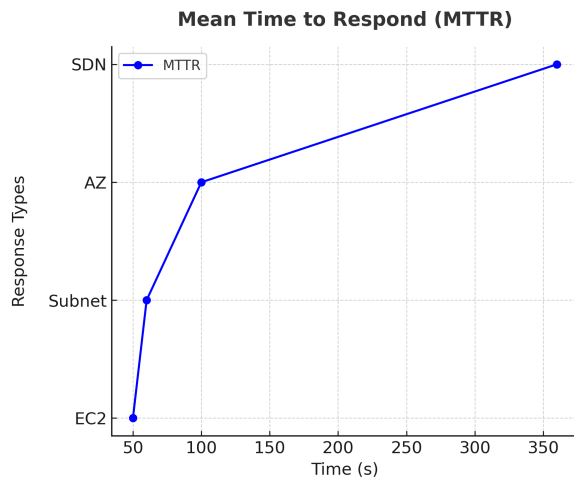


Fig. 2: The Mean Time to Respond (MTTR) metric: time passed from triggering the response until its successful completion.

is recommended for multi-cloud or hybrid environments. Orchestration tools reduce operational overload by automating routine tasks, but they have limited flexibility in adapting to new attack vectors. They depend on predefined workflows and require periodic manual updates.

3) *Autonomous Cyber Defense (ACyD)*: In an infrastructure integrated with Autonomous Cyber Defense (ACyD) agents, the decisions to execute incident responses are driven by the agent's ability to detect abnormal behavior using pre-trained models and threat intelligence databases [12]. This approach relies on the adequate agent's training. The agents operate independently by continuously monitoring the infrastructure for IOC and trigger responses [13]. Using agents ACyD allows for a real-time response without human intervention, making it ideal for large-scale cloud infrastructures or environments with a low tolerance to delay. As a drawback of this approach, we identified the need to regularly re-train the agents in non-production environments to keep them adaptive to new or complex threats. This continuous training could lead to significant costs and introduces some security challenges due to possible adversarial AI attacks. In addition, relying only on autonomous agents will limit human involvement and can cause false positives or inappropriate responses in delicate situations.

4) *Universal decision-making framework*: To implement next-generation cloud incident response, this paper proposes a novel Universal Decision-Making Framework (UDMF) as a methodological contribution to incident response. The UDMF integrates automation and human oversight to ensure efficient, scalable, and context-aware decision-making across diverse operational environments. It is designed to be universally applicable, adapting to various cloud infrastructures and security postures. The framework operates on two distinct layers: Evaluation and Execution.

At the Evaluation layer, agents or orchestration tools automatically assess incidents and determine their severity

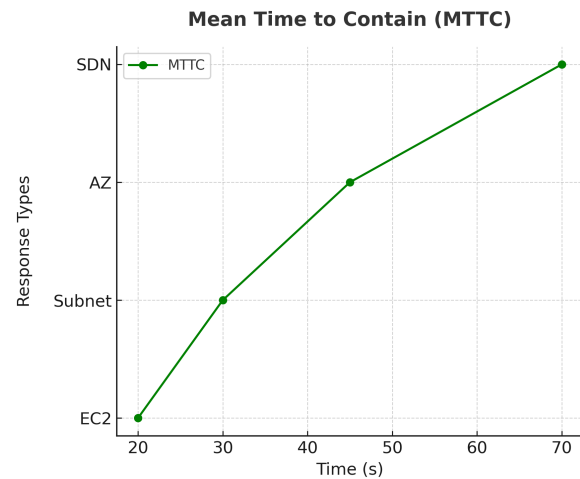


Fig. 3: The Mean Time to Contain (MTTC) metric: time passed from triggering the response to the destruction of the affected instances.

based on received Indicators of Compromise (IoCs). Unlike traditional static assessment models, the UDMF introduces dynamic evaluation mechanisms that adjust severity based on real-time threat intelligence and contextual factors.

At the Execution layer, responses are triggered according to severity: Low- and Medium-severity incidents are handled through automated workflows, ensuring rapid containment.

High-severity incidents are escalated to Security Operations Center (SOC) analysts for further analysis, verification, and strategic decision-making, ensuring that critical threats receive human oversight. The universal nature of this framework allows it to be applied across Public, Private, and Hybrid cloud environments. However, its implementation may require varying levels of resource investment depending on the infrastructure.

Unlike conventional incident response models, which either overrely on automation or depend excessively on manual intervention, the UDMF strikes a balance by combining scalability, adaptability, and human expertise for optimal incident response.

This methodological innovation distinguishes our approach by offering a structured, universally applicable incident response framework. It provides a clear and justifiable rationale for combining automation and human oversight, ensuring that incident responses are both efficient and strategically informed.

B. Response efficiency

Incident responses should be accompanied by performance metrics and sufficient information to analyze results and identify possible future improvements. Effective and immediate incident responses are essential for military networks. Ensuring operational continuity, protecting sensitive data, and ensuring operability through different platforms are critical for decision-makers in these environments. In the following, we discuss several possible metrics that can be used

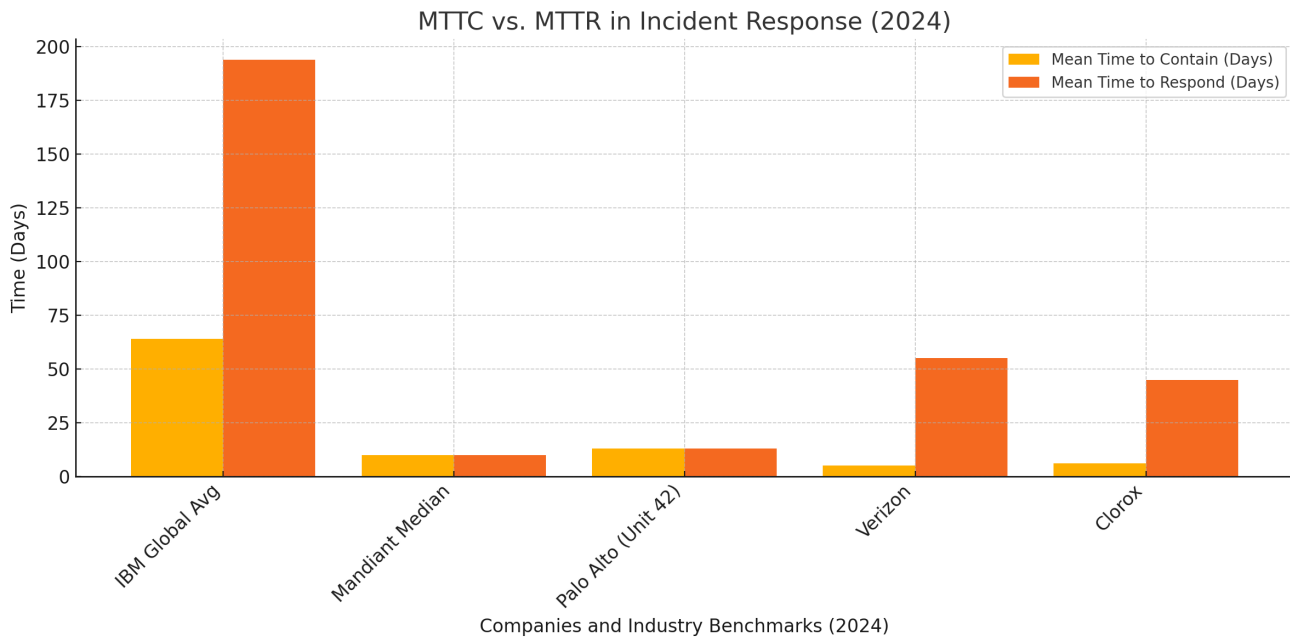


Fig. 4: Industry MTTR and MTTC values in 2024, according to cyber security reports.

to measure the efficiency of responses and find the balance between cost and efficiency.

1) *Response times and KPIs:* Response times are critical in operational networks, where the fast containment of incidents ensures operational continuity and limits adversarial impacts. The proposed multi-layered hierarchical architecture enables efficient response execution across all levels—Instance, Subnet, AZ, and SDN—by utilizing prescribed and automated workflow.

We used two time-related metrics as KPIs to measure the effectiveness of the responses [14]. These are Mean Time to Contain (MTTC) and Mean Time To Respond (MTTR) [15]. According to Cisco [16] and Cloud Security Alliance [5], Mean Time to Contain (MTTC) and Mean Time to Remediate (MTTR) are critical incident response key performance indicators (KPIs). Cisco emphasizes that reducing MTTC and MTTR requires a combination of specialized tools and processes to improve incident containment and remediation efficiency. Similarly, Cloud Security Alliance highlights MTTC as a key metric, as it reflects the overall effectiveness of an incident response team, while MTTR measures the time required to restore systems to an operational state. Both sources underscore the importance of these metrics in evaluating and optimizing incident response strategies.

In our experimental environment, this enveloped the time between triggering the response script and the successful rebuilding of the infrastructure, including the restoration of the normal operations of the affected instances and applications. MTTR represents the time since the incident was detected to its resolution. MTTC, on the other hand, corresponds to the time between incident detection and incident containment. In the context of the described responses, this will include only the time from detecting the

incident to destroying the compromised machines.

Both metrics and the time results we achieved are represented in Fig. 2 and Fig. 3. The values are significantly less than the results reported by industry cyber security reports in 2024 [17], [18]. We examined the 2024 cybersecurity reports from IBM [17], Mandiant [19], Palo Alto Networks [20], Verizon [21], and Clorox [22] to establish a broader understanding of industry benchmarks. Additionally, Fig. 4 provides a comparative analysis of Mean Time to Respond (MTTR) and Mean Time to Contain (MTTC) across different organizations. In this figure, IBM reports the industry average values for MTTR and MTTC, while the other companies present their respective metrics observed during actual security breaches. Both metrics can identify potential areas for improvement within the response scripts and their interaction with the cloud environment. Refining and integrating the responses into the Execution Layer of the Universal Decision Making Framework, mentioned in Section III-B, can enable optimized decision times.

2) *Cost of response:* The cost of response and storage is optimized by employing IaC and automated orchestration tools. By utilizing the on-demand dynamic provisioning of resources in the cloud, and the cost-effective cloud-native tools for the execution of the responses, the system balances cost-efficiency with the robustness required in military information systems infrastructures. The cost of each of the responses reflects the cost of the infrastructure that is affected, e.g., a single instance, multiple instances, or an AZ. Taking snapshots during incident responses ensures not only forensic capabilities but also prevents doubling the cost, as would be the case if we isolate the compromised instances and keep them alive. For the experimental environment, as described in Section II-A the cost is less than 700\$ per

TABLE I: Mapping table between the type of responses and covered MITRE ATT&CK tactics and techniques [23]

Response Type	Tactics Addressed	Techniques Covered
Instance-Level Response	Defense Evasion, Impact	Data Destruction (T1485), Resource Hijacking (T1496), OS Credential Dumping (T1003)
Subnet-Level Response	Lateral Movement, Persistence	Exploitation of Remote Service (T1210), Valid Accounts (T1078), Taint Shared Content (T1080)
AZ-Level Response	Defense Evasion, Command and Control, Impact	Exploitation for Privilege Escalation (T1068), Fallback Channels (T1008), Data Manipulation (T1565)
SDN-Level Response	Command and Control, Exfiltration, Impact	Network Denial of Service (T1498, Exfiltration Over Alternative Protocol (T1048), Service Stop (T1489)

month, with the condition that this is a static experimental environment without inbound or outbound traffic generation.

3) *Critical application availability:* The availability of the mission-critical military application is measured during the execution of incident responses. This metric is used to fine-tune the execution times and the architecture of the reactions. The architecture is designed to restore the availability and accessibility of the applications after an incident response process. By isolating and recreating affected resources in parallel, while rerouting traffic to unaffected components, operational continuity is ensured. Specific strategies are implemented at the SDN responses level to safeguard the infrastructure hosting the critical military application from Denial of Service (DOS) and Distributed Denial of Service (DDOS) attacks. This is achieved during the re-creation actions by dynamically changing the physical location of the EC2s hosting the application to another cloud region and dynamically re-routing the inbound and outbound traffic related to that service, guaranteeing continuity of the critical military application.

4) *CSP-agnostic integration:* The proposed responses are CSP-agnostic, enabling easy integration with different providers or cloud management orchestration tools. By relying on the IaC principles and open CI/CD frameworks, the responses can be deployed in heterogeneous environments, ensuring flexibility and compatibility. This type of adaptability provides readiness across various military cloud and edge infrastructures.

C. Alignment with MITRE ATT&CK matrix

The four types of incident responses described above provide tailored protection and remediation against adversarial activities in a cloud-based infrastructure. These responses align with several tactics and techniques outlined in the MITRE Adversarial Tactics, Techniques, and Common Knowledge (ATT&CK) Matrix [23] for enterprises relevant to cloud systems as shown in Table I and detailed below.

1) *Instance-level response:* Mitigate tactics like Impact Defense Evasion, Impact, and Credential Access quickly remediate compromised resources and effectively wiping the attacker's foothold. This level of response addresses techniques like Data Destruction and Resource Hijacking by eliminating and recreating compromised instances. The response limit the risks of malware persistence and potentially

mitigates risks related to the Credential Dumping technique by isolating instances before attackers can pivot.

2) *Subnet-level response:* Along with covering tactics and techniques from the previous level, this level focuses on Lateral Movement and Persistence tactics within subnet boundaries. Taking a snapshot and redeploying all instances in a compromised subnet limits adversary lateral movement within the network. The subnet-level response extends the scope of remediation by addressing lateral-movement techniques, such as Remote Service Exploitation or Internal Spearfishing.

3) *AZ-level response:* This response adds coverage of Impact, Command and Control(e.g., rerouting network traffic), Privilege Escalation, and Defense Evasion tactics. It can help when subnets or AZs are targeted in distributed attacks. Shifting workloads to an alternate AZ disrupts adversaries' attempts to maintain control over compromised infrastructure.

4) *SDN-level response:* By itself, this response broadly covers Command and Control and Exfiltration tactics. Redeploying infrastructure and reconfiguring routing also mitigates Availability and Impact tactics. It addresses the protection of public-facing applications from Exploitation of Public-Facing applications and Network Denial-of-Service techniques.

D. Different Topologies

The incident responses investigated in our work are intended for cloud environments. However, in addition to the cloud and for the sake of completeness of our research, we compared their implementation and effectiveness in additional topologies, including hybrid cloud and on-premise data centers. Although the characteristics of next-generation cloud incident responses can be adopted for hybrid and on-premise infrastructures, that comes with implementation constraints.

The hybrid cloud includes a combination of on-premise infrastructure and cloud resources, which requires responses to span their actions across all environments. Such a situation will involve orchestration and monitoring tools that can operate simultaneously on-premise and in the cloud. The response scripts must include additional scripting to deal simultaneously with cloud and on-premise virtual snapshots. The responses will have different levels of success in both

environments, which will complicate decision-making and risk management.

In an on-premise environment, responses rely heavily on local resources, limiting scalability. An on-premise implementation attempt of the cloud incident responses will need an orchestration tool that is not cloud-native. Physical infrastructures consist of physical and network access controls that may not exist in the cloud. Implementing the hereby described responses is not recommended in an on-premise infrastructure. It would require re-writing of the response scripts to tailor them to the physical environment and will escape the concepts defining IaC, on which these responses heavily rely.

Although the core principles of next-generation cloud incident responses can be applied across different topologies, their implementation in non-cloud environments is facing constraints and their efficiency decreases, as shown in Table II.

IV. CHALLENGES AND FUTURE RESEARCHES

Introducing incident responses that dynamically change the production infrastructure in a military network is a challenge in itself. This also incorporates difficulties in transforming the traditional approaches to cloud administration and risk management that must be considered. The predefined incident responses could alter the traditional way of cyber defense depending on the risk a defender is willing to take. Such responses could find applications not only in cyber defence, but also on the offensive side of cyber warfare.

A. Reintroducing the incident

One significant challenge in implementing automated responses is the risk of inadvertently reintroducing an incident when re-creating a compromised instance.

When the number of compromised instances within a subnet exceeds a predefined threshold, an orchestration tool or autonomous agent escalates to a higher-level response, executing actions at a broader scale. Integrating predefined scripted responses allows for interaction between cybersecurity engineers and the scripts or the baseline configurations used for the instance re-creation. This approach aims to automate response processes while preserving the critical role of the human factor in cyber security operations. Security engineers can shape the responses to fit the required military decision-making processes better. The responsibility of the human element will remain focused on monitoring the execution of the automated responses, verifying their efficacy, and carrying out further cybersecurity investigations.

B. Drawbacks and Cyber-security concerns

1) *Potential Exploitation by adversaries:* An adversary that gains knowledge of the incident response mechanisms, might deliberately trigger responses in ways that could cause disruptions.

2) *Decision-making risks:* Incident response decisions based on predefined frameworks (such as MITRE ATT&CK) might incorrectly classify events, leading to unnecessary resource destruction or failure to contain an actual attack.

3) *CSP dependency and vendor lock-In:* Certain functionalities provided by the CSP might introduce vendor dependencies that limit portability across different military cloud environments.

4) *Latency and operational continuity challenges:* Large-scale responses (e.g., at the SDN level) might take several minutes, potentially leaving mission-critical applications vulnerable before containment is fully achieved. Rapid destruction and re-creation of instances can introduce temporary service outages, which could affect real-time military operations if not carefully planned.

In order to mitigate the listed drawbacks and cyber-security concerns, a thorough implementation of layered authentication and access controls needs to be in place to prevent unauthorized modifications to automated responses. Introducing the human factor within the mechanisms for high-risk responses can ensure military oversight in critical scenarios. In addition regular monitoring of the responses KPIs should be exercised to optimize response execution times and ensure minimal disruption to mission-critical applications.

C. Risk management

Risk management is one of the few aspects that differ significantly between military and industry-owned cloud infrastructures during cyber security incidents.

Industry-owned infrastructures focus on minimizing downtime, protecting customer data, and mitigating financial losses. The decision-making process is more decentralized, and risk tolerance can vary based on business objectives. In a time of cyber security incidents, companies often have to involve third-party entities, such as insurance companies, external cyber security experts, legal advisors, and CSPs, who influence risk management and decision-making of incident response.

In contrast, in a military cloud infrastructure, risk management often emphasizes national security, operational continuity, and adherence to strict protocols. While industry infrastructures emphasize minimizing downtime cost-effectively to ensure customer satisfaction and financial stability, military infrastructures focus on continuity under extreme conditions, prioritizing mission-critical tasks over cost. Military systems prioritize the resilience and protection of classified data. Often centralized and hierarchical, the decision-making process, especially during cyber incidents, is focused mainly on minimizing the impact on critical military operations and preventing the adversary's access to classified information. It is possible to observe a complete infrastructure shutdown for the sake of damage control, an action that not many industries can afford to take.

Both sectors aim to effectively manage risk in the cloud [24], but their approaches differ in terms of focus, complexity, and specific standards. Industry-owned cloud infrastructures follow cloud risk management standards and frameworks such as International Organization for Standardization (ISO) /International Electrotechnical Commission

TABLE II: Comparison of Responses Efficiency in Different Topologies

Aspect	Cloud	Hybrid	On-Premise
Orchestration Tools	Cloud-native automation	Multi-platform (e.g., Terraform, Kubernetes)	Local tools (e.g., Ansible, Puppet)
Scaling	Dynamic scaling of resources	Limited to available resources in both environments	Fixed capacity based on hardware limits
Snapshots	Cloud snapshots with minimal effort	Mix of cloud and local virtual snapshots	Only local virtual snapshots
Monitoring	Cloud-native monitoring tools	Unified monitoring with cloud and on-prem	On-premise monitoring platforms
Response Speed	High due to automated cloud capabilities	Variable depending on integration	Slower due to manual or limited automation

(IEC) 27001 [25] and 27005 [26], NIST Cyber Security Framework (CSF), Control Objectives for Information and Related Technologies (COBIT) [27], CSA Cloud Control Matrix (CCM) [28], and General Data Protection Regulation (GDPR) [29]. Military networks have to comply with the Alliance standards and frameworks (and in some cases also with national standards), such as the NATO Communications and Information System (CIS) security directive [30], the NATO public cloud security directive [31], European Union (EU). Cybersecurity ACT Framework [32], European Defense Agency (EDA) Framework [33], United States (U.S.) Department of Defense (DoD) Risk Management Framework [34], and NIST SP 800-53 [35].

Developing next-generation cloud incident responses for military networks that can respond quickly to cyber incidents facilitates decision-making by providing an immediate first line of defense and additional time to analyze the impact of the cyber attack. All responses can be adjusted by human operators to suit the constantly changing risk management posture of military networks and align with the requirements of specific operational scenarios.

D. Decoy implementation

In case a threat actor advances quickly through compromised instances and subnets, a possible novel incident response approach could be to dynamically create more and more instances to increase the number of potential targets for the attacker and give cyber defenders time to contain the breach or protect sensitive data. Such an approach could be a last resort in high-stakes military environments where operational continuity and sensitive data preservation are critical. As this approach cannot be used as a standalone solution, it could be a critical component in a layered defense strategy.

E. Honeypot Creation

Honeypot implementations using automated cloud incident responses could add flexibility to the incident response process. In the event of an Indicators of Compromise (IOC) received from multiple instances within a subnet, a modified subnet-level response could be used. It will recreate the affected instances in another subnet to provide operational continuity without destroying the compromised instances. They will be used as a honeypot environment without immediately disrupting the threat actor's activities. This strategy could provide valuable evidence for the

threat actor's Tactics, Techniques, and Procedures (TTPs). Additional measures should be taken to isolate the subnet from the rest of the environment and ensure that sensitive data cannot be exfiltrated from affected machines. Using IaC infrastructure and based on operational requirements, the military cloud infrastructure could also potentially use honeypot environments that intentionally attract sophisticated threat actors. An indication of compromise in such an environment could trigger an automated response that generates additional predefined infrastructure to keep the attacker engaged while analysis of its TTPs is in progress. This ensures that while the attacker keeps control of the original subnet, their actions are limited to interacting with the honeypot systems only.

F. Red team training ground

The ability of automatic responses to use the IaC principles to dynamically change cloud infrastructure can be used for the creation of self-healing or self-generating networks. This could eventually revolutionize red team exercise environments. By using autonomous agents or orchestration tools, such networks can dynamically adapt their topology and conditions in response to the progress of the red team. Upon identifying that a predefined checkpoint or a flag has been captured by the red team, an agent or orchestration tool can autonomously orchestrate the creation of additional network segments and introduce more sophisticated challenges. This adaptive mechanism can progressively increase the difficulty of the exercise, ensuring continuous engagement and improving training efficacy by simulating a more resilient and evolving adversarial environment. This approach could be used in red team exercises with predefined levels of hardness, which measures, e.g., how deep in such a self-generating network a red team can reach in a specific amount of time.

V. CONCLUSIONS

In this paper, we have presented the results of our initial investigation on incident responses in military networks. We have identified and provided a high-level evaluation of several novel approaches to developing next-generation cloud incident response mechanisms adapted to military networks. We have shown how IaC principles and automated orchestration tools can improve resilience, scalability, speed, and efficiency of incident response. The hierarchical approach to responses ensures rapid containment and recovery mechanisms while maintaining the

availability of mission-critical applications. The experimental results reveal significantly better performance in terms of MTTR and MTTC metrics compared to publicly reported benchmarks for industry cloud networks. The responses align well with the adversarial tactics and techniques available from public sources. During our investigation, we analyzed key challenges related to the implementation of response, including the possibility of reintroducing incidents, adapting risk management to military-specific operations, and balancing automation with human oversight. While decoys and honeypots are well-known concepts, the automation of their deployment as part of an incident response strategy represents an innovative approach. Integrating automated decoy-based responses and dynamic honeypot formation offers possibilities for implementing new techniques to incident mitigation and intelligence gathering. Although this innovative way of providing incident responses offers numerous advantages, it also has certain drawbacks and cybersecurity concerns that must be addressed. Enhanced protection should be ensured against vulnerabilities in automation tools, insider threats, and adversarial access to the response mechanism. Cyber-defenders must understand and mitigate these concerns to adapt their responses to emerging threats.

Taking into account recent research in the field of ACyD [11]–[13], [36], [37], our future work will focus on integrating agents as trigger mechanisms for incident responses, aiming to provide a real-world exercise environment for agents' testing and validation. To achieve this, we plan to align our efforts with the IST-162 and IST-196 NATO research groups, with the goal of presenting the results of this future work during NATO 2025 Innovation Continuum cycle [38].

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